THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 03

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** variable we control	Input of the experiment
2	**(d)** can remain liquid	Habitable zone
3	**(b)** carbon dioxide	Inflates the balloon in the activity
4	**(b)** chronic diseases	Long lasting
5	**(c)** removing pathogens	Hygiene
6	**(b)** an iron core	Iron strengthens the field
7	**(c)** zinc container	Container acts as negative
8	**(c)** liquids and gases	Fluids oppose motion
9	**(d)** pressure is largest near the bottom	Horizontal pressure increases with depth
10	**(c)** crouch low in an open area away from tall objects	Minimise contact with ground
11	**(b)** definite volume but no fixed shape	Particles move within space
12	**(c)** iron sulfide	A new compound
13	**(a)** mixture of gases	Nitrogen, oxygen, argon and others
14	**(b)** water	Medium in larger amount
15	**(a)** convex mirror	Gives a wider field of view
16	**(b)** Moon’s phases	Months follow phases
17	**(b)** decrease	Less food available
18	**(a)** soil	Non-living
19	**(a)** life	All living things
20	**(b)** inside the female body	Internal fertilisation

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	A microalga (1). Used as a nutritious food supplement (1).	1+1
22	It avoids illness and suffering (1). It saves cost and prevents spread (1).	1+1
23	Chemicals get used up (1). It cannot be recharged and must be disposed of (1).	1+1
24	Wind from sea to land by day (1). Land heats faster so low pressure forms over land (1).	1+1
25	Hydrogen (1). Iron + dilute HCl → iron chloride + hydrogen (1).	1+1
26	Ice is less dense than water (1). Its particles are farther apart (1).	1+1
27	One benefits, the other is unaffected (1). Orchid on a tree (1).	1+1
28	Ensures continuity of life (1). Without it species would disappear (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Does a puri puff with different flours? (1). How quickly does sugar dissolve in warm and cold water? (1). How does the Moon's shape change over days? (1).	1+1+1
30	Mass in kg, weight in N (1). Mass constant, weight varies (1). Mass is amount of matter, weight is a force (1).	1+1+1
31	Chair is made of matter, which is made of particles (1½). Nothing would remain (1½).	1½+1½
32	They need an enlarged image (1½). Concave mirror gives erect magnified image at short distance (1½).	1½+1½
33	Earth revolves around the Sun (1½). Seasons repeat as Earth moves between equinoxes (1½).	1½+1½
34	Offspring show new features (1½). Variation helps adapt to changing environments (1½).	1½+1½
35	Two cells give more current (1). More current means more heat in filament (1). Hotter filament glows brighter (1).	1+1+1

	OR: Use wires of suitable thickness (1½). Use plugs and sockets rated for the current (1½).	1½+1½
36	Deforestation (1). Pollution (1). Invasive species or overexploitation (1).	1+1+1
	OR: Protect habitats (1½). Conserve biodiversity (1½).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	TB: bacteria affecting lungs, cough, fever (1½). Prevention: avoid close contact, mask, vaccination (1). Hepatitis A: virus affecting liver, jaundice (1½). Prevention: boiled water, vaccination (1).	1½+1+1½+1
	OR: Smallpox had a single stable virus and effective vaccine (1½). Mass vaccination stopped its spread (1). Influenza viruses change often (1). New vaccines are needed (1½).	1½+1+1+1½
38	Pressure differences cause wind (1). Sea and land breezes (1). Storms from low pressure over heated land (1). Cyclones from very low pressure over ocean (1). Speed larger with greater difference (1).	1+1+1+1+1
	OR: Less height gives lower pressure (2). P = F/A (1). 900 ÷ 0.3 (1). 3000 N/m² (1).	2+1+1+1
39	Cylinder on flat surface (1). Eye level with meniscus (1). Read bottom of the curve (1). Balance zero before weighing (1). Record units (1).	1+1+1+1+1
	OR: Salt–water (1). Alcohol–water (1). Soda water (1). Brass (1). Definition of solution (1).	1+1+1+1+1
40	Web diagram with arrows (1). Producer identified (1). Consumers levels named (1). Removal increases mice and frogs (1). Hawks lose a food source and balance changes (1).	1+1+1+1+1
	OR: Clean air and water (1). Food and medicine (1). Climate regulation (1). Human survival depends (1). Conservation protects future (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER